

NATIONAL INSTITUTE OF TECHNOLOGY WARANGAL

Department of Electronics and Communication Engineering

FORMAL-AMBIG-RTL

Phase-Wise Research Working Plan

*Formally Certified Ambiguity Handling in
Large Language Model-Based FSM and RTL Generation*

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Master Research Execution Protocol

The following ten-phase plan defines the complete research workflow from research definition through reproducibility and publication.

Research Workflow Overview

The project is divided into two major scientific stages.



Part I – Build Scientific Ground Truth

Define → Design → Construct → Formally Certify → Human Validate

Part II – Study LLM Behavior

Setup → Experiment → Analyze → Write → Release

1 Phase 1: Research Definition

Goal:

Turn the current idea into a precise, testable research protocol.

1.1 1.1 Deliverables

- Finalized one-paragraph problem statement.
- Precise definition of certified ambiguity.
- Finalized ambiguity taxonomy.
- Definitions of:
 - ASK
 - ASSUME
 - MISINTERPRET
- Formal evaluation plan for single-shot and interactive settings.
- Written benchmark-construction protocol.

1.2 1.2 Certified Ambiguity Definition

The core definition is:

$$CA(S) = P(I_A, I_B \mid S) \wedge (R_A \neq R_B) \quad (1)$$

where:

- S is the natural-language specification.
- I_A and I_B are competing interpretations.

- R_A and R_B are their reference RTL implementations.
- $P(I_A, I_B \mid S)$ represents evidence that both interpretations are plausible readings of the specification.

1.3 1.3 Ambiguity Taxonomy

The initial taxonomy consists of:

1. Reset semantics
2. Output semantics
3. Transition semantics
4. Timing semantics
5. State semantics

1.4 1.4 Response Definitions

ASK:

The model identifies an unresolved behavioral choice and asks a relevant clarification question.

ASSUME:

The model chooses one plausible interpretation without requesting clarification.

MISINTERPRET:

The generated RTL matches neither certified reference interpretation.

1.5 1.5 Exit Criteria

Proceed only when the research protocol has fixed:

1. what constitutes a certified ambiguous instance;
2. what evidence is required to retain or reject an instance;
3. which metrics will be reported;
4. which formal tools will be used;
5. which HDL subset will be supported;
6. which verification assumptions will be used.

Critical principle:

Definitions must be frozen before observing final model results.

2 Phase 2: Benchmark Design

Goal:

Design the benchmark before generating a large number of examples.

2.1 2.1 Benchmark Schema

Each benchmark instance will contain:

Table 1: Benchmark Instance Schema

Field	Contents
Instance ID	Unique identifier, e.g., RESET_001.
Natural-language specification	Intentionally underspecified FSM/RTL requirement.
Category	Reset, output, transition, timing, or state.
Ambiguity dimension	Exact unresolved behavior.
Interpretation A	First plausible reading.
Interpretation B	Second plausible reading.
RTL-A	Reference implementation.
RTL-B	Alternative reference implementation.
Formal environment	Clock, reset, input assumptions, and compared outputs.
Formal result	Equivalent or non-equivalent.
Counterexample	Trace showing behavioral divergence.
Annotation result	Plausibility judgments and agreement.
Final label	Certified ambiguous, unambiguous control, or rejected candidate.

2.2 2.2 Required Supporting Artifacts

- Standard RTL template.
- Formal harness template.
- Hardware-expert annotation form.
- Prompt templates for every LLM condition.
- Category distribution target.

2.3 2.3 Initial Dataset Scale

Table 2: Initial Dataset Targets

Dataset Partition	Target
Candidate ambiguity prompts	80–120
Certified ambiguity cases	40–80
Unambiguous controls	20–40
Negative candidates	20–40

The final certified set does not need to equal the candidate count.

Rejected examples are scientifically useful because they demonstrate that the certification protocol can discriminate between genuine ambiguity and false candidates.

2.4 2.4 Exit Criteria

Proceed when one sample requirement can pass through:

$$\boxed{\textit{Specification} \rightarrow \textit{Paired RTL} \rightarrow \textit{Formal Result} \rightarrow \textit{Annotation} \rightarrow \textit{Final Label}}$$

without changing the protocol.

3 Phase 3: Candidate Construction

Goal:

Build a broad pool of natural-language hardware specifications and paired interpretations.

3.1 3.1 Work Items

1. Create candidate specifications for each ambiguity category.
2. Keep wording natural enough to resemble real engineering requirements.
3. Keep each specification focused on one primary ambiguity whenever possible.
4. Write two interpretations that differ only in the unresolved requirement dimension.
5. Write a short rationale explaining why both interpretations may be reasonable.
6. Create clearly unambiguous control specifications.
7. Create negative candidates that initially appear ambiguous but should be rejected after validation.

3.2 3.2 Example: Reset Ambiguity

Specification:

“When reset is asserted, the controller returns to IDLE.”

Interpretation A:

Reset is synchronous. The FSM enters IDLE at the next active clock edge while reset is asserted.

Interpretation B:

Reset is asynchronous. The FSM immediately enters IDLE when reset is asserted, independent of the clock edge.

Both interpretations should use the same states, inputs, and intended functionality except for reset timing.

3.3 3.3 Candidate Requirements

Every candidate must contain:

1. one natural-language requirement;

2. exactly stated alternative interpretations;
3. a documented ambiguity rationale;
4. a planned observable behavioral difference.

Important:

A candidate must not move forward merely because two different RTL implementations can be written.

The alternatives must be legitimate readings of the original language.

3.4 Exit Criteria

A candidate is ready for Phase 4 only when all four candidate requirements are satisfied.

4 Phase 4: Reference RTL and Formal Certification

Goal:

Convert plausible interpretations into executable reference designs and establish whether their behavior differs.

4.1 Work Items

1. Implement RTL-A and RTL-B.
2. Compile and lint both designs.
3. Synthesize where necessary.
4. Build a common formal wrapper/harness.
5. Define input and reset assumptions explicitly.
6. Define externally observable behavior.
7. Run equivalence or non-equivalence checking.
8. Save a counterexample trace for every non-equivalent pair.
9. Document why the counterexample represents the relevant ambiguity.

4.2 Observable Behavior

The formal environment may compare:

- outputs;
- handshake signals;
- state visibility where appropriate;
- error signals;
- completion signals.

4.3 4.3 Critical Formal Rule

A pair may be marked formally non-equivalent only if the behavioral difference occurs under legal environmental assumptions.

$$\boxed{\textit{Certified NonEquivalence} \Rightarrow \textit{Legal Behavioral Counterexample}} \quad (2)$$

A difference that appears only under an impossible input sequence must not automatically qualify as meaningful ambiguity.

4.4 4.4 Formal Result Classification

Table 3: Formal Certification Decision Table

Result	Action
Formally equivalent	Reject as a certified ambiguity case.
Non-equivalent under legal assumptions	Send for human plausibility validation.
Tool inconclusive	Revise the harness or exclude the case.
RTL invalid	Repair reference RTL before further testing.

4.5 4.5 Exit Criteria

A candidate progresses only if:

1. both RTL implementations compile successfully;
2. the formal environment is documented;
3. formal analysis finds a legal behavioral counterexample;
4. the counterexample is saved and understood.

5 Phase 5: Human Plausibility Validation

Goal:

Establish that non-equivalent RTL pairs represent genuine linguistic ambiguity rather than arbitrary implementation alternatives.

5.1 5.1 Work Items

1. Recruit independent annotators with relevant FSM/RTL experience.
2. Keep annotators independent from RTL authors where possible.
3. Blind interpretation labels.
4. Ask annotators to independently evaluate both interpretations.
5. Measure agreement.
6. Predefine a plausibility threshold.

5.2 5.2 Annotation Questions

Annotators should answer:

1. Is Interpretation A a reasonable reading of the specification?
2. Is Interpretation B a reasonable reading?
3. Does the original specification leave this design choice unresolved?
4. If not, what wording makes one interpretation unreasonable?

5.3 5.3 Agreement Measurement

When multiple annotators are used, inter-annotator agreement may be measured using Fleiss' kappa.

5.4 5.4 Acceptance Rule

Let:

$$P_A = \text{proportion accepting Interpretation A}$$

and

$$P_B = \text{proportion accepting Interpretation B.}$$

Then:

$$\boxed{Certified = FormalNonEquivalent \wedge (P_A \geq \tau) \wedge (P_B \geq \tau)} \quad (3)$$

where τ is a predefined threshold.

A candidate threshold may be:

$$\tau = 0.75$$

or:

$$\tau = 0.80.$$

The final value must be fixed before reviewing final experimental results.

5.5 5.5 Certified Benchmark

At the end of Phase 5:

$$\boxed{\text{FORMAL-AMBIG-RTL Certified Benchmark}}$$

will contain cases satisfying both:

Formal Non-Equivalence

and:

Human Plausibility

6 Phase 6: LLM Evaluation Setup

Goal:

Build a controlled and reproducible evaluation harness.

6.1 Experimental Conditions

Table 4: LLM Evaluation Conditions

Condition	Input to Model	Expected Evaluation
Single-shot	Specification only	ASK, ASSUME, or generate RTL.
Interactive	Specification with clarification permitted	Identify ambiguity, ask useful question, then generate RTL.
Unambiguous control	Fully specified requirement	Generate RTL without needless clarification.
Negative candidate	Candidate that failed certification	Avoid treating all vague wording as automatically ambiguous.

6.2 Model Configuration

Document:

- exact model name;
- model version;
- access date;
- fixed prompt template;
- temperature;
- maximum output tokens;
- sampling parameters;
- random seed where supported.

6.3 Required Model Output

The requested response structure should contain:

1. identified ambiguity, if any;
2. clarification question, if any;

3. assumption, if used;
4. complete synthesizable RTL.

6.4 6.4 Interactive Trials

Controlled designer answers should be created for interactive trials.

The experiment should balance answers selecting Interpretation A and Interpretation B.

6.5 6.5 Logging

Log:

- prompts;
- responses;
- timestamps;
- token usage;
- latency;
- generated RTL.

6.6 6.6 Exit Criteria

The pilot setup must verify that:

1. all model responses are captured;
2. RTL extraction works reliably;
3. generated RTL can be compiled;
4. formal analysis can be executed;
5. all question/response logs are preserved.

No human should manually modify generated RTL without recording the intervention as a separate experimental condition.

7 Phase 7: Experimental Execution

Goal:

Run the complete benchmark across selected models and conditions.

7.1 7.1 Work Items

1. Evaluate every selected model on every certified ambiguity.
2. Evaluate the same models on unambiguous controls.
3. Evaluate negative candidates.

4. Run single-shot and interactive conditions.
5. Use repeated trials where generation is non-deterministic.
6. Compile generated RTL.
7. Formally compare generated RTL with RTL-A and RTL-B.

7.2 7.2 RTL Classification

Generated RTL will be classified as:

$$Class(R_{LLM}) = \begin{cases} A, & R_{LLM} \equiv R_A \\ B, & R_{LLM} \equiv R_B \\ AB, & R_{LLM} \equiv R_A \wedge R_{LLM} \equiv R_B \\ M, & \text{otherwise} \end{cases} \quad (4)$$

An additional category will be used:

Invalid

when parsing, elaboration, compilation, or formal evaluation fails.

7.3 7.3 Behavioral Classification

LLM responses will additionally be classified as:

$ASK \quad ASSUME \quad MISINTERPRET$

8 Phase 8: Analysis and Results

Goal:

Answer the research questions with evidence.

8.1 8.1 Core Analysis

Table 5: Research Question to Evidence Mapping

Research Question	Main Evidence
Can models recognize certified ambiguity?	Ambiguity Detection Rate and clarification recall.
Do models ask useful questions?	Clarification precision, recall and question-quality score.
Which ambiguity categories are hardest?	Per-category ASK, ASSUME, MISINTERPRET and invalid rates.
Does interaction help?	Paired single-shot versus interactive formal-correctness comparison.
Do models produce unsupported behavior?	Misinterpretation rate and counterexample analysis.
What is the cost of clarification?	Tokens, turns, latency and clarification utility.

8.2 8.2 Ambiguity Detection Rate

$$ADR = \frac{N_{\text{correctly detected}}}{N_{\text{certified ambiguous}}} \quad (5)$$

8.3 8.3 Clarification Rate

$$CR = \frac{N_{\text{relevant ASK}}}{N_{\text{certified ambiguous}}} \quad (6)$$

8.4 8.4 Valid Interpretation Rate

$$VIR = \frac{N_A + N_B}{N_{\text{generated}}} \quad (7)$$

8.5 8.5 Misinterpretation Rate

$$MIR = \frac{N_M}{N_{\text{generated}}} \quad (8)$$

8.6 8.6 Formal Correctness Rate

$$FCR = \frac{N_{\text{formally correct}}}{N_{\text{attempted generations}}} \quad (9)$$

8.7 8.7 Clarification Utility

$$CU = \frac{FCR_{\text{interactive}} - FCR_{\text{single-shot}}}{C_{\text{interaction}}} \quad (10)$$

where $C_{\text{interaction}}$ may include:

- number of turns;
- token usage;

- latency;
- or another predefined interaction-cost measure.

8.8 Unnecessary Clarification Rate

For unambiguous controls:

$$UCR = \frac{N_{\text{ASK on unambiguous controls}}}{N_{\text{unambiguous controls}}} \quad (11)$$

This prevents a model from appearing strong simply by asking for clarification on every specification.

8.9 Exit Criteria

Before moving to final writing, produce:

- main result tables;
- confidence intervals;
- category-wise breakdowns;
- single-shot versus interactive comparisons;
- representative formal counterexamples;
- clear answers to every research question.

9 Results-Freeze Protocol

The complete dataset should contain one row per:

$$(Model, Benchmark Instance, Condition, Trial) \quad (12)$$

After the complete dataset is collected:

$$Results Frozen \rightarrow Statistical Analysis \rightarrow Paper Conclusions$$

After freezing:

- definitions should not be changed;
- metrics should not be retroactively redefined;
- cases should not be selectively removed;
- conclusions must follow the collected evidence.

10 Phase 9: Paper Completion

Goal:

Transform the completed experiment into a publication-quality empirical paper.

10.1 9.1 Required Paper Updates

Proposal statements such as:

- “expected results”;
- “TBD”;
- “no experimental results are claimed”

must be replaced by actual results after experimentation.

10.2 9.2 Required Final Evidence

The final paper should report:

- benchmark certification yield;
- annotation agreement;
- plausibility threshold results;
- formal-tool configuration;
- formal proof outcomes;
- model-specific results;
- statistical tests;
- confidence intervals;
- formal counterexample case studies;
- artifact availability.

10.3 9.3 Recommended Contribution Structure

The final contribution list may contain:

1. A hardware-specific benchmark for evaluating ambiguity handling in natural-language-to-FSM/RTL generation.
2. A certification methodology combining independent human plausibility validation with formal non-equivalence checking.
3. A curated set of certified ambiguous, unambiguous, and rejected/negative-control specifications.
4. Paired reference RTL and formal verification artifacts.
5. A formal behavioral classification method distinguishing interpretation selection from misinterpretation.
6. An empirical evaluation of single-shot and interactive LLM-based RTL generation.

11 Phase 10: Reproducibility and Submission

Goal:

Make the research reviewable, reproducible, and reusable.

11.1 10.1 Repository Structure

```

FORMAL-AMBIG-RTL/
|
+-- benchmark/
|   +-- specifications/
|   +-- interpretations/
|   +-- controls/
|   +-- annotation_results/
|
+-- rtl/
|   +-- reference_A/
|   +-- reference_B/
|
+-- formal/
|   +-- harnesses/
|   +-- assumptions/
|   +-- scripts/
|   +-- counterexamples/
|
+-- prompts/
|
+-- model_outputs/
|
+-- evaluation/
|   +-- rtl_extraction/
|   +-- classification/
|   +-- metrics/
|   +-- statistics/
|
+-- results/
|
+-- paper/
|
+-- README.md

```

11.2 10.2 Final Submission Checklist

1. Benchmark instances have stable IDs.
2. Reference RTL and formal scripts reproduce reported outcomes.
3. Counterexamples are retained.
4. Annotation methodology is documented.
5. Agreement scores are reported.

6. Model names and versions are documented.
7. Prompts and decoding parameters are disclosed.
8. Raw model outputs are preserved.
9. Undisclosed manual modification of model RTL is prohibited.
10. Data and code licensing are documented.
11. Every quantitative claim is backed by an experiment.
12. Every table contains final values.
13. References are bibliographically verified.
14. Venue-specific formatting and originality requirements are satisfied.

12 Master Phase Timeline

Table 6: Master Research Execution Timeline

Period	Phase	Primary Output
Late Sep–Oct 2026	Phase 1: Research Definition	Frozen problem statement, definitions, taxonomy, metrics and protocol.
Oct 2026	Phase 2: Benchmark Design	Benchmark schema, RTL template, formal harness and annotation forms.
Oct–Nov 2026	Phase 3: Candidate Construction	80–120 candidate specifications.
Nov–Dec 2026	Phase 4: Formal Certification	Formally verified candidate pool.
Dec 2026	Phase 5: Human Validation	40–80 certified ambiguity cases plus controls.
Jan 2027	Phase 6: LLM Evaluation Setup	Controlled and reproducible evaluation harness.
Jan–Feb 2027	Phase 7: Experimental Execution	Complete model/instance/condition/trial dataset.
Feb 2027	Phase 8: Analysis and Results	Metrics, statistical analysis and case studies.
Mar 2027	Phase 9: Paper Completion	Final manuscript and results.
Mar–Apr 2027	Phase 10: Reproducibility and Submission	Artifact package and paper submission.
2027	Thesis Integration	M.Tech thesis and final defense.

13 Phase Dependency Map

The research dependencies are:

14 Critical Research Gates

Table 7: Mandatory Research Gates

Gate	Stage	Pass Condition
G1	After Phase 1	Definitions, taxonomy and metrics frozen.
G2	After Phase 2	One complete benchmark instance can pass through the protocol.
G3	After Phase 4	Legal formal counterexample establishes behavioral difference.
G4	After Phase 5	Both interpretations satisfy predefined plausibility threshold.
G5	After Phase 6	LLM responses and generated RTL are captured automatically.
G6	After Phase 7	Complete experimental dataset is available.
G7	After Phase 8	Every research question has evidence.
G8	Before submission	Paper, benchmark and reproducibility artifacts are complete.

15 Final Research Principle

The complete project can be summarized as:

$$\begin{array}{l}
 \text{Define} \rightarrow \text{Design} \rightarrow \text{Construct} \\
 \rightarrow \text{Formally Certify} \rightarrow \text{Human Validate} \\
 \rightarrow \text{Evaluate LLMs} \rightarrow \text{Analyze} \\
 \rightarrow \text{Publish} \rightarrow \text{Release}
 \end{array} \tag{13}$$

The central methodological principle is:

**Build the formal ground truth first.
Measure the LLM second.**

This ordering ensures that the LLM experiment is based on a scientifically validated benchmark rather than on assumptions made after observing model behavior.

Conclusion

The ten-phase plan provides a complete research execution framework for FORMAL-AMBIG-RTL.

Phases 1–5 establish the scientific ground truth:

$$\text{Define} \rightarrow \text{Design} \rightarrow \text{Construct} \rightarrow \text{Formal Certification} \rightarrow \text{Human Validation}$$

Phases 6–10 use that ground truth to evaluate LLMs and transform the results into a reproducible publication:

LLM Setup → Experiment → Analysis → Paper → Reproducibility

The plan therefore treats the project as a sequence of scientific gates rather than as a simple implementation schedule.